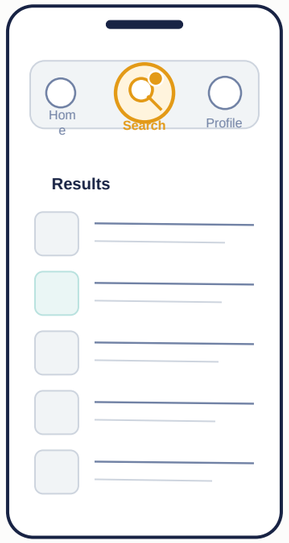
Interaction-Graph Action Mapping Across UI Layouts

Map Canonical Nodes through unified visual–layout evidence and iterative contextual correspondence.

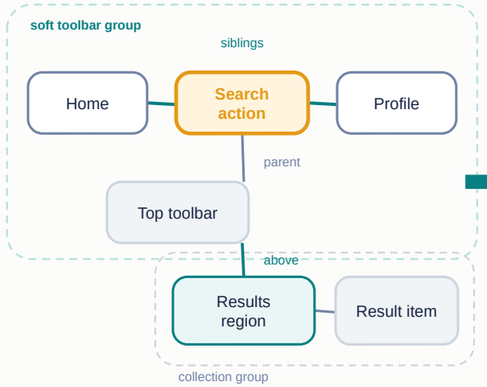
RECORDED OBSERVATION + ACTION



Recorded tap

Search is in the top toolbar

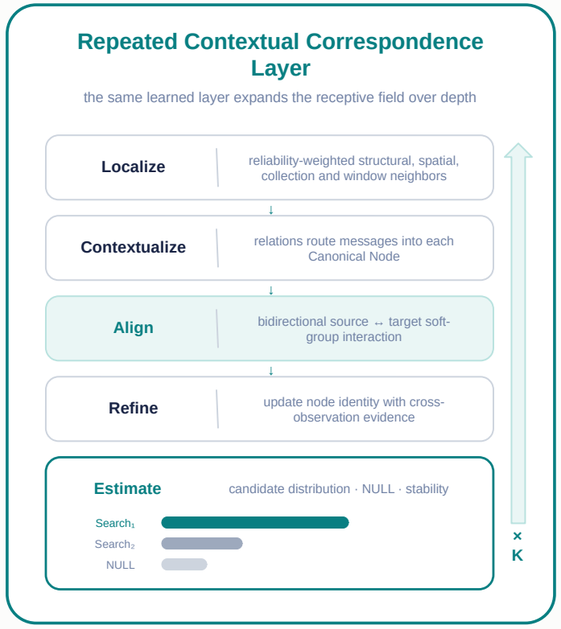
SOURCE OBSERVATION GRAPH



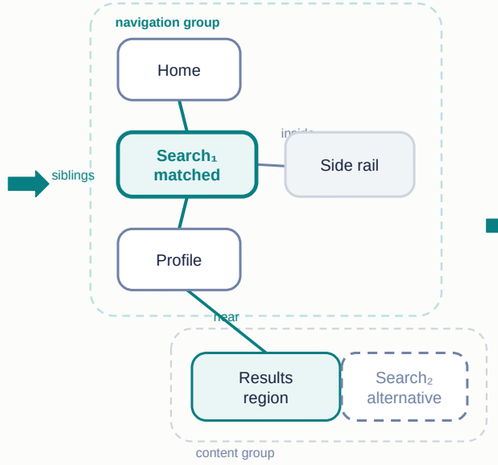
Canonical action node + Observation Graph

semantic · visual internal/ring/group · role/state
geometry · reliability · group membership

CONTEXTUAL CORRESPONDENCE ×K



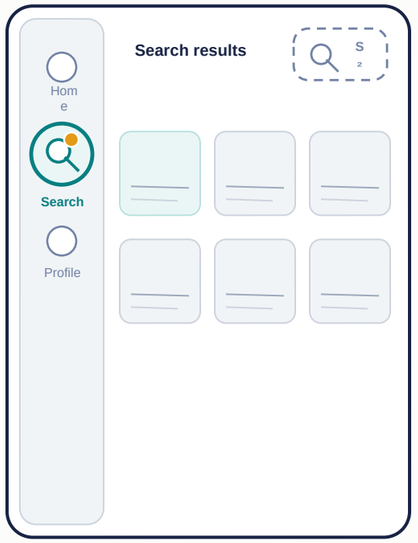
TARGET OBSERVATION GRAPH



Canonical target candidates + soft groups

groups may overlap; attention can escape a wrong group

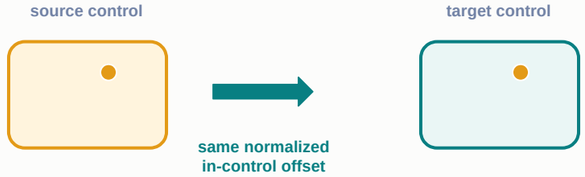
MAPPED TARGET ACTION



Mapped tap

Search moved to the side rail; a similar icon remains in content

RELATIVE ACTION PROJECTION



$\delta = (p_s - b_s^{tl}) / \text{size}(b_s)$

$p_t = b_t^{tl} + \delta \odot \text{size}(b_t)$

No source-coordinate replay

Why the mapping survives layout change

Each Canonical Node fuses semantic, multi-scale visual, role/state, geometry, reliability and group evidence. Structural, spatial, collection and window relations route messages inside repeated correspondence layers; they are not added later as a hand-written score.

LEGEND ○ recorded action ○ mapped target ⬚ alternative candidate